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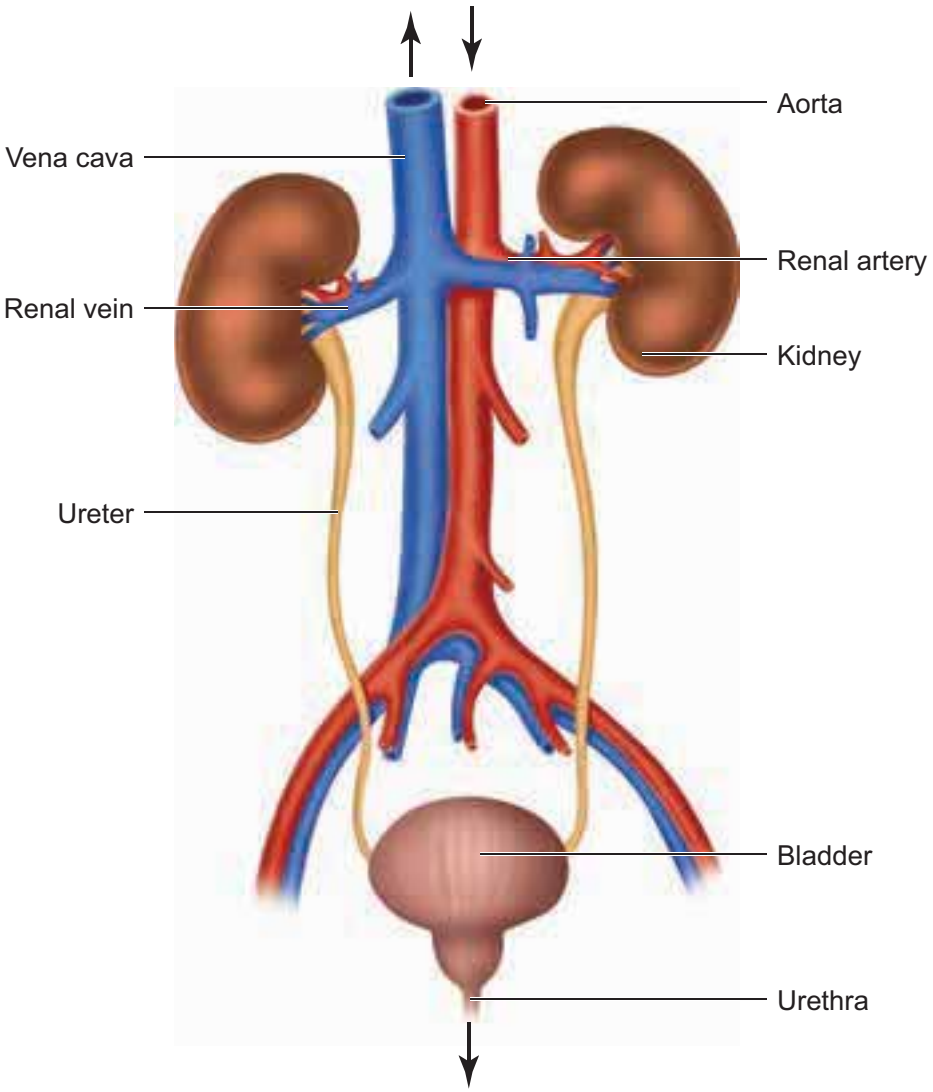

Figure 8.2 Human excretory system

8.2 Human excretory system

8.2.1 Structure of kidney

Excretory system in human consists of a pair of kidneys, a pair of ureters, urinary bladder and urethra (Figure. 8.2). Kidneys are reddish brown, bean shaped structures that lie in the superior lumbar region between the levels of the last thoracic and third lumber vertebra close to the dorsal inner wall of the abdominal cavity. The right kidney is placed slightly lower than the left kidney. Each kidney weighs an average of 120-170 grams. The outer layer of the kidney is covered by three layers of supportive tissues namely, renal fascia, perirenal fat capsule and fibrous capsule.

The longitudinal section of kidney (Figure. 8.3) shows, an outer cortex, inner medulla and pelvis. The medulla is divided into a few conical tissue masses called medullary pyramids or renal pyramids. The part of cortex that extends in between the medullary pyramids is the renal columns of **Bertini.** The centre of the inner concave surface of the kidney has a notch called the renal **hilum,** through which ureter, blood vessels and nerves innervate. Inner to the hilum is a broad funnel shaped space called the renal pelvis with projection called calyces. The